

TalkTrace Database Schema Specification

Complete Relational Data Architecture & Key Constraints

TABLE 1: users (System User Accounts)

Stores core account information for all system users across all roles (admins, subscribers, host users). Manages unique login credentials, email verification status, and security parameters.

No	Column Name	Data Type	Key / Constraint	Description
1	uid	VARCHAR(128)	PK	Unique Firebase / Auth User Id...
2	name	VARCHAR(100)	NOT NULL	Full Name of User
3	email	VARCHAR(150)	UNIQUE, NOT NULL	Login Email Address
4	emailVerified	BOOLEAN	DEFAULT FALSE	Email Verification Confirmatio...
5	role	ENUM(admin,user)	DEFAULT user	Access Control Role
6	createdAt	TIMESTAMP	DEFAULT NOW()	Account Creation Date

TABLE 2: meetings (Meeting Sessions & Audio Meta)

Holds metadata for recorded meeting sessions, including language auto-detection, audio duration, processing lifecycle state, and owner association.

No	Column Name	Data Type	Key / Constraint	Description
1	id	VARCHAR(128)	PK	Unique Meeting UUID
2	user_id	VARCHAR(128)	FK (users.uid)	Owner User Identifier
3	bot_id	VARCHAR(128)	NULLABLE	Recall.ai Note-Taker Bot ID
4	status	VARCHAR(50)	DEFAULT processing	Overall Status (processing, co...
5	process_state	VARCHAR(50)	DEFAULT initializing	Detailed Pipeline State (trans...
6	language	VARCHAR(10)	DEFAULT auto	Google / Whisper Verified Lang...
7	duration_seconds	INTEGER	DEFAULT 0	Audio Duration in Seconds
8	created_at	BIGINT	NOT NULL	Session Epoch Timestamp (ms)

TABLE 3: meeting_keys (Zero-Trust Cryptographic Key Vault)

Encrypted vault storing AES-256 keys and initialization vectors (IV) for local zero-trust audio and transcript encryption.

No	Column Name	Data Type	Key / Constraint	Description
1	meeting_id	VARCHAR(128)	PK, FK (meetings.id)	Associated Meeting ID
2	encrypted_key_blob	TEXT	NOT NULL	Protected Key Vault Blob (AES/...

TABLE 4: transcript_revisions (Version History & Hash Audit Log)

Tracks version history and SHA-256 cryptographic hashes for transcripts and executive summaries to guarantee audit immutability.

No	Column Name	Data Type	Key / Constraint	Description
1	id	INTEGER	PK, AUTOINCREMENT	Revision Record ID
2	meeting_id	VARCHAR(128)	FK (meetings.id)	Associated Meeting ID
3	version	INTEGER	NOT NULL	Revision Version Number (1, 2,...
4	type	ENUM(transcript, summary)	NOT NULL	Revision Artifact Type
5	content_hash	VARCHAR(64)	NOT NULL	SHA-256 Cryptographic Hash
6	file_path	VARCHAR(255)	NOT NULL	Encrypted Vault File Path
7	edited_at	BIGINT	NOT NULL	Revision Edit Timestamp (ms)

TABLE 5: speaker_profiles (Recurring Speaker Voice & Name Registry)

Stores persistent participant voice profiles and custom name mappings across meetings for automated Pyannote speaker recognition.

No	Column Name	Data Type	Key / Constraint	Description
1	id	INTEGER	PK, AUTOINCREMENT	Profile Record ID
2	user_id	VARCHAR(128)	FK (users.uid)	Owner User ID
3	speaker_label	VARCHAR(50)	NOT NULL	Pyannote Diarization Label
4	speaker_name	VARCHAR(100)	NOT NULL	Real Human Name (e.g. Abin Geo...
5	voice_signature	TEXT	NULLABLE	Dense Voice Embedding Vector
6	updated_at	BIGINT	NOT NULL	Last Update Timestamp (ms)

TABLE 6: action_items (AI Action Items & Task Manager Board)

Contains structured tasks automatically extracted by local LLMs (Ollama) from meeting transcripts, with priorities.

No	Column Name	Data Type	Key / Constraint	Description
1	id	VARCHAR(128)	PK	Action Item Unique ID
2	meeting_id	VARCHAR(128)	FK (meetings.id)	Originating Meeting ID
3	task	TEXT	NOT NULL	Action Item Task Description
4	assignee	VARCHAR(100)	DEFAULT Unassigned	Responsible Participant
5	deadline	VARCHAR(50)	DEFAULT ASAP	Targeted Due Date / Timeframe
6	priority	ENUM(High,Medium,Low)	DEFAULT Medium	Task Urgency Rating
7	completed	BOOLEAN	DEFAULT FALSE	Task Completion Status
8	confidence	FLOAT	DEFAULT 0.95	LLM Extraction Confidence Score

TABLE 7: **calendar_events (Google & Outlook Synced Meetings)**

Stores synced calendar events from Google Calendar and Microsoft Outlook Graph APIs for automated bot scheduling.

No	Column Name	Data Type	Key / Constraint	Description
1	id	VARCHAR(128)	PK	Calendar Event Unique ID
2	user_id	VARCHAR(128)	FK (users.uid)	Associated User Account
3	title	VARCHAR(255)	NOT NULL	Meeting Title / Subject
4	start_time	TIMESTAMP	NOT NULL	Scheduled Start DateTime
5	meeting_link	TEXT	NOT NULL	Google Meet / Teams / Zoom URL
6	source	ENUM(Google, Outlook)	NOT NULL	Calendar Source API
7	bot_scheduled	BOOLEAN	DEFAULT FALSE	Auto-Join Bot Scheduled Flag

TABLE 8: **search_history (Natural Language Vector RAG Query Logs)**

Stores natural language search queries and 128-dimensional sentence embedding vector logs for client-side search audit.

No	Column Name	Data Type	Key / Constraint	Description
1	id	INTEGER	PK, AUTOINCREMENT	Log ID
2	user_id	VARCHAR(128)	FK (users.uid)	Query User ID
3	query	TEXT	NOT NULL	Natural Language Search String
4	created_at	BIGINT	NOT NULL	Search Timestamp (ms)